

Parathyroid Hormone Physiology

Calcium Homeostasis · PTH · Calcitonin · Calcitriol

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PHS 220 — Physiology

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§1 PARATHYROID GLAND ANATOMY

The parathyroid glands are four small, oval endocrine glands (each approximately $6 \times 4 \times 2$ mm, weighing ~40 mg) embedded in the posterior surface of the thyroid gland — two superior and two inferior. Despite their small size, they are critical for life: complete removal (e.g., inadvertently during total thyroidectomy) causes fatal hypocalcaemic tetany within days.

Feature	Description
Number	Usually 4 (occasionally 3–6); supernumerary glands may be found in ectopic locations (thymus, mediastinum, retrooesophageal)
Location	Posterior surface of thyroid: superior pair at junction of upper/middle thirds; inferior pair near lower pole
Blood supply	Inferior thyroid arteries (primarily); very important to preserve during thyroidectomy
Embryology	Superior pair from 4th pharyngeal pouch; Inferior pair from 3rd pharyngeal pouch (with thymus)
Chief cells	Main functional cells; synthesise and secrete PTH; contain pale granular cytoplasm
Oxyphil cells	Function unclear; increase after puberty; may be reserve/exhausted chief cells

1.1 Parathyroid Hormone (PTH) — Structure & Synthesis

Property	Detail
Chemical class	Peptide hormone — single chain polypeptide of 84 amino acids
Synthesis pathway	Pre-pro-PTH (115 aa) → Pro-PTH (90 aa) → PTH (84 aa) — enzymatic cleavage in ER and Golgi
Storage	Stored as prohormone in secretory granules; rapid release on demand
Active fragment	N-terminal region PTH(1-34) contains full biological activity
Half-life	~2–4 minutes (rapidly cleared by liver and kidney)
Receptor	PTH1R — Class B GPCR; primarily $G_s \rightarrow cAMP$; also activates $G_q \rightarrow IP_3/Ca^{2+}$
Second messenger	cAMP (predominant); IP_3/DAG in some tissues
Solubility	Hydrophilic peptide; travels free in plasma

PTH SECRETION STIMULI — KEY VALUES

Primary stimulus: Hypocalcaemia (ionised Ca^{2+} below ~ 1.1 mmol/L).

The parathyroid chief cell directly senses ionised Ca^{2+} via the **Calcium-Sensing Receptor (CaSR)** — a cell-surface GPCR. When ionised Ca^{2+} falls $\rightarrow$ CaSR activity decreases $\rightarrow$ PTH secretion increases (inverse relationship).

Additional stimuli for PTH release: Low Mg^{2+} (hypomagnesaemia), high phosphate.

Inhibitors of PTH release: High Ca^{2+} (hypercalcaemia); High Mg^{2+} (hypermagnesaemia); Calcitriol (vitamin D — long-loop negative feedback on parathyroid).

§2 IMPORTANCE OF CALCIUM

Calcium is the most abundant mineral in the body (~ 1 kg in adults, 99% in bone as hydroxyapatite). In the extracellular fluid, calcium concentration is maintained within an extremely narrow range because small deviations cause profound physiological dysfunction.

2.1 Normal Values & Forms of Calcium

Form	% of Total	Details
Ionised (free) Ca^{2+}	$\sim 50\%$	Physiologically active form; measured directly; affected by pH (acidosis $\uparrow$, alkalosis $\downarrow$)
Protein-bound	$\sim 41\%$	Mainly albumin-bound ($\sim 80\%$ of protein-bound); also globulins. NOT physiologically active. Reduced by hypoalbuminaemia.
Complexed	$\sim 9\%$	Bound to anions (citrate, phosphate, sulphate, oxalate). Not immediately active.

CORRECTED CALCIUM FORMULA

Total serum calcium is affected by albumin concentration. If albumin is low, total calcium appears falsely low.

Corrected Ca = Measured total Ca (mmol/L) + $0.02 \times (40 - \text{albumin g/L})$

Normal total calcium: 2.1–2.6 mmol/L (8.4–10.4 mg/dL).

Normal ionised Ca^{2+} : 1.0–1.3 mmol/L (4.0–5.2 mg/dL).

2.2 Functions of Calcium

Function	Mechanism / Relevance
Structural (bone & teeth)	Hydroxyapatite $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$ provides rigidity and strength; bone is also the body's Ca^{2+} reservoir
Muscle contraction	Ca^{2+} binds troponin-C (skeletal/cardiac) or calmodulin-MLCK (smooth muscle) $\rightarrow$ actin-myosin cross-bridge cycling
Neurotransmitter release	Action potential $\rightarrow$ voltage-gated Ca^{2+} channels open $\rightarrow$ Ca^{2+} influx $\rightarrow$ synaptic vesicle exocytosis

Function	Mechanism / Relevance
Cardiac pacemaking	Ca ²⁺ responsible for phase 2 (plateau) of cardiac action potential; ICa(L) through L-type channels
Enzyme cofactor	Essential for coagulation (factors II, VII, IX, X require Ca ²⁺ for activity)
Second messenger	IP ₃ releases Ca ²⁺ from ER → activates calmodulin-dependent kinases, PKC
Membrane excitability	Extracellular Ca ²⁺ stabilises Na ⁺ channel threshold; hypocalcaemia → ↓ threshold → tetany
Hormone secretion	Required for exocytosis of secretory granules (insulin, PTH, etc.)

§3 CALCIUM HOMEOSTASIS — OVERVIEW

Serum calcium is maintained by a tightly coordinated interplay of three principal hormones acting on three target organs (bone, kidney, intestine). The central principle is that ionised Ca²⁺ is sensed by the CaSR on parathyroid cells and the C-cells of the thyroid, triggering reciprocal adjustments in PTH and calcitonin secretion.

Hormone	Source	Stimulus	Net Effect on Serum Ca ²⁺	Works Synergistically With
PTH	Parathyroid chief cells	Hypocalcaemia; ↓ Mg ²⁺ ; ↑ phosphate	↑ (raises Ca ²⁺)	Calcitriol (both raise Ca ²⁺)
Calcitriol (1,25-OH ₂ -D ₃)	Kidney (1α-hydroxylase, stimulated by PTH)	↓ Ca ²⁺ ; ↓ phosphate; PTH; ↓ FGF-23	↑ (raises Ca ²⁺)	PTH
Calcitonin	Thyroid C-cells (parafollicular)	Hypercalcaemia; gastrin	↓ (lowers Ca ²⁺)	Antagonises PTH & calcitriol
Oestrogen / Prolactin / GH	Ovary / Pituitary	Oestrogen: low levels at menopause	Indirect: stimulate 1α-hydroxylase → ↑ Ca ²⁺ absorption during pregnancy, lactation, growth	—

§4 ACTIONS OF PARATHYROID HORMONE (PTH)

PTH acts on three principal target organs — bone, kidney, and intestine (indirectly via calcitriol) — to raise serum calcium while lowering serum phosphate. All direct actions are mediated through the PTH1 receptor coupled to cAMP (and to a lesser extent IP₃/Ca²⁺).

4.1 Actions on Bone

Acute (minutes-hours): PTH promotes rapid Ca²⁺ efflux from superficial bone surfaces (osteocytic osteolysis), providing an immediate buffer for acute hypocalcaemia.

Chronic (sustained PTH elevation) — Bone Resorption:

- PTH binds PTH1R on osteoblasts (osteoclasts do not directly express PTH receptors).
- Activated osteoblasts secrete RANK Ligand (RANKL) — a cytokine that binds RANK receptors on osteoclast precursors.
- RANKL stimulates osteoclast differentiation and activation. (OPG — osteoprotegerin — is a decoy receptor for RANKL that antagonises this effect.)
- Activated osteoclasts secrete cathepsin K and acid phosphatases → dissolve collagen matrix and hydroxyapatite → release Ca^{2+} and PO_4^{3-} into circulation.
- Net result: Bone resorption → ↑ serum Ca^{2+} and ↑ serum PO_4^{3-} (but see renal phosphate excretion below).

INTERMITTENT vs CONTINUOUS PTH

Intermittent (pulsatile) PTH → ANABOLIC to bone — stimulates osteoblasts > osteoclasts → bone formation. Basis of teriparatide (recombinant PTH 1-34) therapy for osteoporosis.

Continuous (sustained) PTH → CATABOLIC to bone — promotes osteoclast-driven resorption. This is what occurs in primary hyperparathyroidism → 'osteitis fibrosa cystica' in severe disease.

4.2 Actions on the Kidney

Renal Site	PTH Action	Effect
Proximal tubule (PCT)	Inhibits Na-phosphate co-transporter (NaPi-2a)	↑ Phosphaturia (phosphate wasting in urine) → ↓ serum PO_4^{3-}
Loop of Henle (TAL)	Inhibits phosphate reabsorption	Further phosphaturia
Distal tubule (DCT) / Collecting duct	Stimulates Ca^{2+} reabsorption via TRPV5 channels	↑ Ca^{2+} retained → ↑ serum Ca^{2+}
PCT (1 α -hydroxylase)	Upregulates 1 α -hydroxylase enzyme	↑ Calcitriol synthesis → indirect ↑ intestinal Ca^{2+} absorption

4.3 Actions on the Intestine (Indirect)

PTH has no direct action on the intestine. Its indirect effect is via stimulation of calcitriol synthesis in the kidney. Calcitriol then acts on intestinal enterocytes to upregulate calbindin (Ca^{2+} -binding protein) and Ca^{2+} ATPase, significantly increasing transcellular calcium absorption from the gut lumen.

4.4 Net Effect of PTH

Parameter	Change	Mechanism
Serum Ca^{2+}	↑ Increased	Bone resorption + renal Ca^{2+} reabsorption + intestinal absorption (via calcitriol)

Parameter	Change	Mechanism
Serum PO_4^{3-}	↓ Decreased	Phosphaturia (despite phosphate released from bone, renal excretion dominates)
Urinary Ca^{2+}	Paradoxically ↑ in hyperparathyroidism	Filtered load overwhelms tubular reabsorption capacity; despite ↑ tubular reabsorption per nephron
Serum Calcitriol	↑ Increased	PTH stimulates 1α -hydroxylase
Serum Alkaline Phosphatase	↑ in chronic ↑ PTH	Reflects ↑ osteoblast activity (bone turnover marker)

§5 ACTIONS OF CALCITONIN

Calcitonin is a 32-amino acid peptide hormone produced and secreted by the parafollicular cells (C-cells) of the thyroid gland in response to hypercalcaemia. Its mechanism of action is via cAMP. Calcitonin acts as the physiological 'brake' on calcium elevation, directly opposing PTH and calcitriol.

Target Organ	Calcitonin Action	Effect on Serum Ca^{2+}
Bone	Inhibits osteoclast activity and recruitment; stimulates osteoblasts (bone deposition)	↓ (reduces Ca^{2+} release from bone)
Kidney	Signals kidneys to reabsorb less Ca^{2+} (increases urinary Ca^{2+} excretion)	↓ (increases Ca^{2+} loss in urine)
Intestine	Reduces Ca^{2+} absorption from gut	↓ (less Ca^{2+} entering blood from diet)

CLINICAL NOTE ON CALCITONIN

Calcitonin plays a relatively minor role in adult calcium homeostasis (total thyroidectomy rarely causes significant hypercalcaemia). Its importance is greatest in: (1) **Foetal and neonatal life** — protects the developing skeleton; (2) **Pregnancy and lactation** — protects maternal bone from calcium depletion; (3) **Pharmacological use** — salmon calcitonin is used to treat hypercalcaemia (Paget's disease, malignancy-related hypercalcaemia) and osteoporosis. **Medullary thyroid carcinoma (MTC)** arises from C-cells and is a calcitonin-secreting tumour — elevated calcitonin is a tumour marker.

§6 ACTIONS OF CALCITRIOL (ACTIVE VITAMIN D)

6.1 Synthesis Pathway

Step	Location	Conversion	Enzyme / Factor
1	Skin	7-dehydrocholesterol → Cholecalciferol (Vitamin D3)	UV-B radiation (sunlight)
2	Liver	Cholecalciferol → 25-hydroxycholecalciferol (25-OH-D3 / Calcidiol)	25-hydroxylase
3	Kidney	25-OH-D3 → 1,25-dihydroxycholecalciferol (Calcitriol — active form)	1 α -hydroxylase (stimulated by PTH, ↓ Ca ²⁺ , ↓ PO ₄ ³⁻)
Inactive pathway	Kidney	25-OH-D3 → 24,25-dihydroxycholecalciferol (inactive)	24-hydroxylase (dominant when Ca ²⁺ is high; inactivation pathway)

6.2 Target Organ Actions of Calcitriol

Calcitriol acts via the nuclear Vitamin D Receptor (VDR), which heterodimerises with RXR to bind Vitamin D Response Elements (VDREs) and regulate gene transcription.

Target Organ	Actions of Calcitriol	Net Effect on Ca ²⁺ /PO ₄ ³⁻
Intestine (primary)	↑ Calbindin-D9k synthesis → active transcellular Ca ²⁺ absorption; ↑ TRPV6, Ca ²⁺ -ATPase, Na/Ca ²⁺ antiporter; also ↑ phosphate absorption via NaPi	↑ Ca ²⁺ and PO ₄ ³⁻ absorbed from diet
Kidney	↑ Ca ²⁺ reabsorption in DCT (↑ calbindin-D28k); ↓ renal Ca ²⁺ excretion	↑ serum Ca ²⁺
Bone	Direct: increases membrane permeability to Ca ²⁺ ; stimulates RANKL → osteoclast activation → bone resorption. Also promotes osteoblast differentiation.	↑ serum Ca ²⁺ and PO ₄ ³⁻ (from bone)
Parathyroid glands	Negative feedback: downregulates PTH gene transcription; reduces parathyroid cell proliferation	↓ PTH → indirect Ca ²⁺ regulation

Net Effect of Calcitriol: ↑ serum Ca²⁺ and ↑ serum PO₄³⁻ (unlike PTH which lowers phosphate).

§7 DISORDERS OF CALCIUM REGULATION

7.1 Hypercalcaemia (serum Ca²⁺ > 2.6 mmol/L)

Cause	Mechanism	Key Features	Lab Clue
Primary hyperparathyroidism	Parathyroid adenoma (85%), hyperplasia (15%), carcinoma (<1%) → autonomous PTH excess	Often asymptomatic; Stones (renal calculi), Bones (pain, subperiosteal resorption), Groans (GI: N/V, constipation), Psychic moans (depression, cognitive)	High PTH; High Ca^{2+} ; Low PO_4^{3-} ; High urinary Ca^{2+}
Malignancy-related	PTHrP (PTH-related protein) from tumour (humoral hypercalcaemia); osteolytic metastases; calcitriol (lymphoma)	Weight loss; systemic malignancy symptoms	Low/undetectable PTH; High Ca^{2+} ; High PTHrP
Vitamin D toxicity	Excess exogenous calcitriol or granulomatous disease (sarcoid, TB) → ↑ 1α -hydroxylase	Fatigue, polyuria, polydipsia, nephrocalcinosis	Low PTH; High calcitriol (25-OH-D3 in exogenous toxicity; 1,25-OH ₂ -D3 in granulomatous)
Milk-Alkali syndrome	Excess Ca^{2+} and absorbable alkali intake	Metabolic alkalosis; renal impairment	Normal/low PTH; metabolic alkalosis

SYMPTOMS OF HYPERCALCAEMIA — Mnemonic: **Bones, Stones, Groans, Psychic Moans**

Bones: Bone pain, pathological fractures, osteitis fibrosa cystica (severe chronic ↑ PTH).

Stones: Renal calculi (calcium oxalate/phosphate); nephrocalcinosis.

Groans: Nausea, vomiting, constipation, anorexia, pancreatitis.

Psychic moans: Depression, confusion, cognitive impairment; coma in severe cases.

Cardiac: Shortened QT interval on ECG; bradycardia; hypertension.

7.2 Hypocalcaemia (serum Ca^{2+} < 2.1 mmol/L)

Cause	Mechanism	Key Features
Hypoparathyroidism	Surgical (post-thyroidectomy most common); autoimmune (DiGeorge/22q11); infiltration	Tetany; perioral paraesthesias; Chvostek's sign; Trousseau's sign; prolonged QTc; cataracts
Vitamin D deficiency	Poor diet/sunlight; malabsorption; liver/kidney disease	Rickets (children); osteomalacia (adults); muscle weakness; bone pain
Pseudohypoparathyroidism	Target organ resistance to PTH (GNAS mutation)	Features of hypoparathyroidism but HIGH PTH; Albright's hereditary osteodystrophy (short stature, short 4th metacarpal)
Hypomagnesaemia	Mg^{2+} required for PTH secretion and action; alcoholism, malabsorption	Refractory hypocalcaemia until Mg^{2+} corrected
Acute pancreatitis	Ca^{2+} saponification by peripancreatic fat necrosis	Severe pancreatitis; indicator of severity (poor prognosis)

CLINICAL SIGNS OF HYPOCALCAEMIA

Chvostek's sign: Tapping over the facial nerve (just anterior to the ear) → ipsilateral facial muscle twitching (upper lip, cheek).

Trousseau's sign: Inflating a BP cuff above systolic for 3 minutes → carpal spasm (main d'accoucheur position — wrist flexion, MCP extension, thumb adduction).

Tetany: Spontaneous painful muscle cramps/spasms — particularly hands, feet, face.

Due to ↓ membrane stabilisation by Ca^{2+} → ↓ Na^+ channel threshold → spontaneous firing.

ECG: Prolonged QTc interval (risk of Torsades de Pointes).